

CSA 821: Machine Learning

Assignment 1: Development of a Machine Learning Models for Real-World Data Problems

Instructions:

1. You may work individually or in groups of up to five students.
2. Select any real-world dataset from sources such as:
Kaggle, UCI Machine Learning Repository or any other source of your convenience.
3. Data Preprocessing
 - Handle missing values and outliers appropriately.
 - Perform data standardization or normalization where applicable.
 - Conduct feature selection or dimensionality reduction if necessary.
 - Perform encoding using one hot encoding where necessary
 - Provide summary statistics and visualizations to describe the dataset.
4. Model Development
 - Implement at least two (2) different machine learning algorithms, e.g. Logistic Regression, Decision Trees, Random Forest, SVM, XGBoost, KNN, etc depending on whether your output is discrete or continuous.
 - Split your data into training and test sets.
 - Train each model and compare their performance.
5. Model Optimization
 - Apply hyperparameter tuning techniques such as: GridSearchCV or RandomizedSearchCV etc
 - Discuss the chosen hyperparameters and the rationale for selection.
6. Performance Evaluation
 - Use appropriate evaluation metrics depending on your task:
 - Classification: Accuracy, Precision, Recall, F1-score, ROC-AUC
 - Regression: RMSE, MAE, R^2
 - Present results in tables and visualizations (e.g., confusion matrix, ROC curves).
7. Submission
 1. Submit your work as either: A Python script (.py), or A Jupyter Notebook (.ipynb) with well-commented code and clear label of the group members names and registration number at the beginning of the code. If assignment is done in groups, each member MUST submit the work individually in SOMAS
8. Bonus Marks (up to +5%)
 - Using a dataset relevant to the Kenyan context (e.g., agriculture, healthcare, education, transport, or fintech).
 - Use of data visualization libraries such as Matplotlib, Seaborn, or Plotly to communicate insights effectively.